

## Year 7 · Chapter 2 · Angles: answers (teacher copy)

### 2.01 Naming angles (booklet p. 3)

#### Examples (model these live)

E1 Name the marked angle.

vertex in the middle ·  $\angle PQR$  or  $\angle RQP$

E2 Name the vertex and the arms of  $\angle XYZ$ .

vertex: Y · arms: YX and YZ

E3 Name the three angles at O.

small:  $\angle AOB$ ,  $\angle BOC$  · whole:  $\angle AOC$

#### Questions (31 on today's skill, 10 from earlier lessons)

- |                                |                                                             |                                                                                                       |                                                                                                       |
|--------------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 1 U                            | 13 $\angle DMB$ or $\angle BMD$                             | 23 $16^{1.06}$                                                                                        | 33 $= 9 - 23 = -14^{1.08}$                                                                            |
| 2 $\angle YQC$ or $\angle CQY$ | 14 $\angle ECX$                                             | 24 $\angle HVW$ and $\angle WVG$                                                                      | 34 $21^\circ + 26^\circ = 47^\circ$                                                                   |
| 3 $\angle VPW$                 | 15 vertex: H · arms: HC and HB                              | 25 $= 12 - 17 = -5^{1.04}$                                                                            | 35 6 angles · $\angle PJC$ , $\angle CJY$ , $\angle YJZ$ · $\angle PJY$ , $\angle CJZ$ , $\angle PJZ$ |
| 4 E                            | 16 small: $\angle RTQ$ , $\angle QTB$ · whole: $\angle RTB$ | 26 vertex: N · arms: NS and NX                                                                        | 36 6 angles · $\angle AOK$ , $\angle KOW$ , $\angle WOG$ · $\angle AOW$ , $\angle KOG$ , $\angle AOG$ |
| 5 $\angle BZA$ or $\angle AZB$ | 17 $-1^{1.02}$                                              | 27 $42^\circ + 52^\circ = 94^\circ$                                                                   | 37 No. They have different vertices (H and Q).                                                        |
| 6 $\angle LPG$                 | 18 CP                                                       | 28 $13^{1.05}$                                                                                        | 38 $6^{1.02}$                                                                                         |
| 7 V                            | 19 $\angle UBL$ and $\angle LBF$                            | 29 $46^\circ + 49^\circ = 95^\circ$                                                                   | 39 $60^\circ + 27^\circ = 87^\circ$                                                                   |
| 8 $\angle AZM$ or $\angle MZA$ | 20 $(( - ) 22) \times 2 = \cdot = 484^{1.09}$               | 30 6 angles · $\angle COW$ , $\angle WOG$ , $\angle GOU$ · $\angle COG$ , $\angle WOU$ , $\angle COU$ | 40 bracket: $-168 \cdot = 2\ 184^{1.09}$                                                              |
| 9 $\angle NKX$                 | 21 small: $\angle MDS$ , $\angle SDN$ · whole: $\angle MDN$ | 31 $5^{1.03}$                                                                                         | 41 $\angle ZVR$ and $\angle RVE$                                                                      |
| 10 M                           | 22 RN                                                       | 32 No. They have different vertices (N and E).                                                        |                                                                                                       |
| 11 $26^{1.01}$                 |                                                             |                                                                                                       |                                                                                                       |
| 12 vertex: S · arms: SD and SU |                                                             |                                                                                                       |                                                                                                       |

### 2.02 Measuring and drawing angles (booklet p. 5)

#### Examples (model these live)

E1 Measure the angle.

$0^\circ$  on the bottom arm ·  $= 55^\circ$

E2 Draw an angle of  $130^\circ$  on the ray.

centre on the end,  $0^\circ$  on the ray · mark  $130^\circ$ , rule the arm

E3 How do you draw a reflex angle of  $250^\circ$ ?

$360^\circ - 250^\circ = 110^\circ$  · draw  $110^\circ$ , mark the outside

#### Questions (27 on today's skill, 8 from earlier lessons)

- |                                                       |                                                                    |                                                                                  |                                                                                 |
|-------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 1 $155^\circ$                                         | 13 estimate: about $110^\circ$ · measure: $114^\circ$              | 21 small: $\angle THP$ , $\angle PHS$ · whole: $\angle THS^{2.01}$               | 29 $360^\circ - 30^\circ = 330^\circ$ · $330^\circ$ is reflex                   |
| 2 $360^\circ - 200^\circ = 160^\circ$                 | 14 $45^\circ$ (check with a protractor)                            | 22 $360^\circ - 230^\circ = 130^\circ$ · draw $130^\circ$ , mark the outside     | 30 each number: $360^\circ \div 12 = 30^\circ$ · $2 \times 30^\circ = 60^\circ$ |
| 3 more                                                | 15 $\angle UTA$ and $\angle ATZ^{2.01}$                            | 23 $360^\circ - 70^\circ = 290^\circ$ · $290^\circ$ is reflex                    | 31 10 apart, half is 5 · 5 right of -10: $-5^{1.02}$                            |
| 4 $25^\circ$                                          | 16 inside angle: $75^\circ$ · $360^\circ - 75^\circ = 285^\circ$   | 24 $\angle MQK$ and $\angle KQA^{2.01}$                                          | 32 Read the wrong scale. The angle is $65^\circ$ .                              |
| 5 $360^\circ - 225^\circ = 135^\circ$                 | 17 $135^\circ$ (check with a protractor)                           | 25 each number: $360^\circ \div 12 = 30^\circ$ · $6 \times 30^\circ = 180^\circ$ | 33 $\angle MWB$ or $\angle BWM^{2.01}$                                          |
| 6 less                                                | 18 inside angle: $165^\circ$ · $360^\circ - 165^\circ = 195^\circ$ | 26 $4^{1.03}$                                                                    | 34 $360^\circ - 205^\circ = 155^\circ$ · draw $155^\circ$ , mark the outside    |
| 7 $360^\circ - 290^\circ = 70^\circ$                  | 19 $\angle EFZ$ or $\angle ZFE^{2.01}$                             | 27 Read the wrong scale. The angle is $25^\circ$ .                               | 35 inside angle: $160^\circ$ · $360^\circ - 160^\circ = 200^\circ$              |
| 8 more                                                | 20 estimate: about $130^\circ$ · measure: $131^\circ$              | 28 $360^\circ - 220^\circ = 140^\circ$ · draw $140^\circ$ , mark the outside     |                                                                                 |
| 9 $20^\circ$                                          |                                                                    |                                                                                  |                                                                                 |
| 10 $61 - 69 = -8$ dollars $1.10$                      |                                                                    |                                                                                  |                                                                                 |
| 11 estimate: about $110^\circ$ · measure: $109^\circ$ |                                                                    |                                                                                  |                                                                                 |
| 12 $360^\circ - 320^\circ = 40^\circ$                 |                                                                    |                                                                                  |                                                                                 |

### 2.03 Classifying angles (booklet p. 7)

#### Examples (model these live)

E1 Classify  $135^\circ$ .

between  $90^\circ$  and  $180^\circ$  · obtuse

E2 Classify  $270^\circ$ .

between  $180^\circ$  and  $360^\circ$  · reflex

E3 Classify the angle between the clock hands at 4:00.

$4 \times 30^\circ = 120^\circ$  · obtuse

#### Questions (36 on today's skill, 13 from earlier lessons)

- |                                        |                                                                           |                                                            |                                                                                |
|----------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------|--------------------------------------------------------------------------------|
| 1 reflex                               | 11 acute                                                                  | 21 $360^\circ - 210^\circ = 150^\circ$ · obtuse            | 29 $35^\circ + 165^\circ = 200^\circ$ · reflex                                 |
| 2 acute                                | 12 $-1^{1.04}$                                                            | 22 $1^{1.03}$                                              | 30 Yes. Each is more than $90^\circ$ , so the total is more than $180^\circ$ . |
| 3 between $90^\circ$ and $180^\circ$   | 13 $15^\circ + 45^\circ = 60^\circ$ · acute                               | 23 $15^\circ + 10^\circ = 25^\circ$ · acute                | 31 total: $-20 \cdot = -20 \div 4 \cdot = -5^\circ C^{1.07}$                   |
| 4 reflex                               | 14 obtuse                                                                 | 24 $4 \times 30^\circ = 120^\circ$ · obtuse                | 32 $4 \times 30^\circ = 120^\circ$ · obtuse                                    |
| 5 acute                                | 15 revolution                                                             | 25 $90^\circ + 60^\circ = 150^\circ$ · obtuse              | 33 No. The obtuse angle alone is more than $90^\circ$ .                        |
| 6 between $0^\circ$ and $90^\circ$     | 16 $\angle ULV$ or $\angle VLU^{2.01}$                                    | 26 $\angle FPV^{2.01}$                                     | 34 $5 \times 45^\circ = 225^\circ$ · reflex                                    |
| 7 straight                             | 17 $30^\circ + 75^\circ = 105^\circ$ · obtuse                             | 27 $360^\circ - 205^\circ = 155^\circ$ · obtuse            |                                                                                |
| 8 right                                | 18 $2 \times 30^\circ = 60^\circ$ · acute                                 | 28 estimate: about $40^\circ$ · measure: $44^\circ^{2.02}$ |                                                                                |
| 9 acute                                | 19 inside angle: $120^\circ$ · $360^\circ - 120^\circ = 240^\circ^{2.02}$ |                                                            |                                                                                |
| 10 between $180^\circ$ and $360^\circ$ | 20 $180^\circ + 85^\circ = 265^\circ$ · reflex                            |                                                            |                                                                                |

- 35  $160^\circ$  (check with a protractor) <sup>2.02</sup>  
 36  $360^\circ \div 6 = 60^\circ$  · acute  
 37  $= 36 - 6 \cdot = 30$  <sup>1.08</sup>  
 38 small angle:  $90^\circ \cdot 360^\circ - 90^\circ = 270^\circ$   
 39 No. Each is less than  $90^\circ$ , so the total is less than  $180^\circ$ .  
 40  $\angle VAP$  or  $\angle PAV$  <sup>2.01</sup>  
 41  $1.5 \times 90^\circ = 135^\circ$  · obtuse  
 42  $360^\circ \div 4 = 90^\circ$  · right  
 43  $(-22) \times^2 = \cdot = 484$  <sup>1.09</sup>  
 44 small angle:  $30^\circ \cdot 360^\circ - 30^\circ = 330^\circ$   
 45  $\angle SBW$  <sup>2.01</sup>  
 46  $360^\circ - 310^\circ = 50^\circ$  · acute  
 47  $3 \times 60^\circ = 180^\circ$  · straight  
 48 estimate: about  $40^\circ$  · measure:  $41^\circ$  <sup>2.02</sup>  
 49  $360^\circ \div 8 = 45^\circ$  · acute

## 2.04 Complementary and supplementary angles (booklet p. 9)

Examples (model these live)

- E1 Find the complement of  $35^\circ$ .  
 $90^\circ - 35^\circ \cdot = 55^\circ$   
 E2 Find the supplement of  $112^\circ$ .  
 $180^\circ - 112^\circ \cdot = 68^\circ$   
 E3 Find x.  
 $3x + 60 = 180 \cdot 3x = 120 \cdot x = 40$

Questions (33 on today's skill, 11 from earlier lessons)

- 1  $24^\circ$   
 2  $110^\circ$   
 3 no (they add to  $100^\circ$ )  
 4  $a = 85$   
 5  $27^\circ$   
 6  $62^\circ$   
 7 no (they add to  $100^\circ$ )  
 8  $a = 150$   
 9  $71^\circ$   
 10 no (they add to  $100^\circ$ )  
 11  $a = 30$   
 12  $47 - 105 \cdot = -58$  dollars  
<sup>1.10</sup>  
 13  $54^\circ$   
 14  $m + 73 = 90 \cdot m = 17$   
 15  $58^\circ$   
 16  $360^\circ - 325^\circ \cdot = 35^\circ$  <sup>2.02</sup>  
 17  $y + 50 = 90 \cdot y = 40$   
 18  $x + 79 = 180 \cdot x = 101$   
 19 small:  $\angle GSX, \angle XSR$  · whole:  $\angle GSR$  <sup>2.01</sup>  
 20  $b + 30 = 90 \cdot b = 60$   
 21 Yes.  $149 + 31 = 180$ .  
 22  $3$  <sup>1.03</sup>  
 23  $x + 51 = 180 \cdot x = 129$   
 24  $b + 40 = 90 \cdot b = 50$   
 25 between  $180^\circ$  and  $360^\circ$   
<sup>2.03</sup>  
 26 Yes.  $157 + 23 = 180$ .  
 27 inside angle:  $155^\circ \cdot 360^\circ - 155^\circ = 205^\circ$  <sup>2.02</sup>  
 28  $y + 56 = 90 \cdot y = 34$   
 29  $3x + 30 = 180 \cdot 3x = 150 \cdot x = 50$   
 30 total:  $0 \cdot = 0 \div 3 \cdot = 0^\circ C$  <sup>1.07</sup>  
 31  $3x + 42 = 180 \cdot 3x = 138 \cdot x = 46$   
 32  $x + x + 10 = 90 \cdot 2x = 80 \cdot x = 40$   
 33  $360^\circ - 320^\circ = 40^\circ$  · acute  
<sup>2.03</sup>  
 34  $x = 2(90 - x) \cdot 3x = 180 \cdot x = 60^\circ$   
 35  $21$  <sup>1.01</sup>  
 36 complement:  $65^\circ$  · supplement:  $180^\circ - 65^\circ = 115^\circ$   
 37  $3x + 72 = 180 \cdot 3x = 108 \cdot x = 36$   
 38  $x + x + 20 = 90 \cdot 2x = 70 \cdot x = 35$   
 39  $x = 3(180 - x) \cdot 4x = 540 \cdot x = 135^\circ$   
 40  $= 17 - 22 \cdot = -5$  <sup>1.04</sup>  
 41 complement:  $23^\circ$  · supplement:  $180^\circ - 23^\circ = 157^\circ$   
 42 obtuse <sup>2.03</sup>  
 43  $3x + 102 = 180 \cdot 3x = 78 \cdot x = 26$   
 44 Yes.  $132 + 48 = 180$ .

## 2.05 Angles at a point and vertically opposite angles (booklet p. 11)

Examples (model these live)

- E1 Find a.  
 $a + 125 = 180 \cdot a = 55$   
 E2 Find b.  
 $b + 130 + 110 = 360 \cdot b = 120$   
 E3 Find c. Give a reason.  
 $c = 70 \cdot$  vertically opposite angles

Questions (29 on today's skill, 11 from earlier lessons)

- 1  $a = 150$   
 2  $b = 75$   
 3  $c = 360 - 85 - 95 \cdot c = 180$   
 4  $a = 70$   
 5  $b = 135$   
 6  $c = 360 - 65 - 40 \cdot c = 255$   
 7  $b = 100$   
 8  $c = 360 - 45 - 50 \cdot c = 265$   
 9  $a = 80$   
 10  $6$  <sup>1.01</sup>  
 11  $a + 115 + 100 = 360 \cdot a = 145$   
 12  $b = 55$   
 13  $a + 55 + 85 = 360 \cdot a = 220$   
 14  $c + 55 + 40 = 180 \cdot c = 85$   
 15 small:  $\angle MZK, \angle KZB$  · whole:  $\angle MZB$  <sup>2.01</sup>  
 16  $b = 50$  (vert. opp.) ·  $a = 180 - 50 = 130 \cdot c = 130$  (vert. opp.)  
 17  $c + 25 + 35 = 180 \cdot c = 120$   
 18  $b = 120$  (vert. opp.) ·  $a = 180 - 120 = 60 \cdot c = 60$  (vert. opp.)  
 19  $-1$  <sup>632</sup> <sup>1.09</sup>  
 20  $a + 70 + 95 = 360 \cdot a = 195$   
 21  $\angle BGN$  <sup>2.01</sup>  
 22  $6x = 360 \cdot x = 60$   
 23  $c + 30 + 25 = 180 \cdot c = 125$   
 24 yes <sup>2.04</sup>  
 25  $9x = 360 \cdot x = 40$   
 26  $2x + 15 = 83 \cdot 2x = 68 \cdot x = 34$   
 27  $7 - (-1) \cdot = 7 + 1 \cdot = 8^\circ C$  <sup>1.05</sup>  
 28  $4x + 132 = 180 \cdot 4x = 48 \cdot x = 12$   
 29 more <sup>2.02</sup>  
 30 No. Angles on a straight line add to  $180^\circ$ , so it is  $130^\circ$ .  
 31  $9x = 360 \cdot x = 40$   
 32  $\angle NSY$  and  $\angle YSL$  <sup>2.01</sup>  
 33  $2x + 15 = 53 \cdot 2x = 38 \cdot x = 19$   
 34  $4x + 100 = 180 \cdot 4x = 80 \cdot x = 20$   
 35  $48$  <sup>1.06</sup>  
 36 No. Angles on a straight line add to  $180^\circ$ , so it is  $140^\circ$ .  
 37  $\angle GSK$  or  $\angle KSG$  <sup>2.01</sup>  
 38  $4x = 360 \cdot x = 90$   
 39  $b = 60$  (vert. opp.) ·  $a = 180 - 60 = 120 \cdot c = 120$  (vert. opp.)  
 40 right <sup>2.03</sup>

## 2.06 Parallel and perpendicular lines (booklet p. 13)

Examples (model these live)

- E1 Write with a symbol: AB is perpendicular to CD.  
 $\perp$  means perpendicular ·  $AB \perp CD$   
 E2 Name the parallel sides of rectangle PQRS.  
 opposite sides ·  $PQ \parallel SR, PS \parallel QR$   
 E3 Construct a line perpendicular to AB through P, a point on AB.  
 arcs on AB either side of P · equal arcs cross; rule from P

Questions (31 on today's skill, 10 from earlier lessons)

- 1  $HX \perp WY$   
 2 parallel  
 3 LC is parallel to TV  
 4  $HB \perp DZ$   
 5 perpendicular  
 6 SE is perpendicular to WC  
 7  $CR \perp WD$   
 8 neither  
 9  $b = 95$  <sup>2.05</sup>  
 10 parallel  
 11 TL is parallel to XN  
 12  $PU \perp GD$   
 13  $-465$  <sup>1.09</sup>  
 14  $\parallel: AB \parallel DC, AD \parallel BC \cdot \perp:$  every corner, e.g.  $AB \perp BC$   
 15 perpendicular  
 16 CD is parallel to HK  
 17  $142^\circ$  <sup>2.04</sup>  
 18  $\parallel: PQ \parallel SR, PS \parallel QR \cdot \perp:$  none  
 19  $90^\circ$   
 20 small:  $\angle HSE, \angle ESG$  · whole:  $\angle HSG$  <sup>2.01</sup>  
 21 two lines that meet at  $90^\circ$ , marked with a right-angle mark  
 22  $90^\circ$   
 23  $\parallel: EF \parallel HG \cdot \perp:$  none

24  $= -8 - 6 = -14$  <sup>1.08</sup>  
 25 compasses at V: mark two points on PQ · from each point, equal arcs that cross · rule from V through the crossing  
 26  $360^\circ - 230^\circ = 130^\circ$  <sup>2.02</sup>  
 27  $90^\circ$   
 28  $\parallel$ : WX  $\parallel$  ZY, WZ  $\parallel$  XY ·  $\perp$ : none

29 a line at  $90^\circ$  to the ray through its end point (check with a set square)  
 30 rule a line through V that cuts PQ · copy the angle it makes at V · rule the new arm: it is  $\parallel$  to PQ  
 31 total:  $0 = 0 \div 4 = 0^\circ$  <sup>1.07</sup>  
 32  $AB \perp EF$

33 between  $180^\circ$  and  $360^\circ$  <sup>2.03</sup>  
 34 compasses more than half of PQ · arcs from P and from Q, above and below · rule through the two crossings  
 35  $\perp$  to WX: WZ and XY ·  $\parallel$  to WX: ZY  
 36  $AB \parallel EF$

37  $\perp$  to AB: AD and BC ·  $\parallel$  to AB: DC  
 38 0 <sup>1.05</sup>  
 39  $AB \parallel EF$   
 40 less <sup>2.02</sup>  
 41  $\perp$  to PQ: PS and QR ·  $\parallel$  to PQ: SR

## 2.07 Corresponding angles (booklet p. 15)

### Examples (model these live)

E1 The lines are parallel. Find a.  
 corresponding angles are equal ·  $a = 70$

E2 Find b, the angle next to a on the line.  
 $b + 70 = 180$  (straight line) ·  $b = 110$

E3 Corresponding angles are  $(2x + 10)^\circ$  and  $70^\circ$ . Find x.  
 $2x + 10 = 70$  ·  $2x = 60$  ·  $x = 30$

### Questions (29 on today's skill, 9 from earlier lessons)

1  $a = 135$   
 2  $140^\circ$   
 3 q  
 4  $a = 35$   
 5  $24^\circ$   
 6  $48^\circ$   
 7 p  
 8  $a = 135$   
 9  $-1\ 131$  <sup>1.09</sup>  
 10  $a = 140$  (corresponding) ·  $b = 180 - 140 = 40$   
 11  $44^\circ$   
 12  $a = 40$  (corresponding) ·  $b = 180 - 40 = 140$

13  $a = 150$  · corresponding angles,  $\parallel$  lines  
 14 small:  $\angle UYF, \angle FYC$  · whole:  $\angle UYC$  <sup>2.01</sup>  
 15  $x + 5 = 25$  ·  $x = 20$   
 16  $a = 145$  (corresponding) ·  $b = 180 - 145 = 35$   
 17  $a = 135$  · corresponding angles,  $\parallel$  lines  
 18  $x + 5 = 10$  ·  $x = 5$   
 19  $= -24 - 10 = -34$  <sup>1.08</sup>  
 20  $a = 90$  (corresponding) ·  $b = 180 - 90 = 90$   
 21  $360^\circ - 350^\circ = 10^\circ$  <sup>2.02</sup>  
 22  $4x + 5 = 49$  ·  $4x = 44$  ·  $x = 11$

23  $a = 60$  · corresponding angles,  $\parallel$  lines  
 24  $4x + 5 = 73$  ·  $4x = 68$  ·  $x = 17$   
 25  $y = 180 - 55 = 125$  (straight) ·  $x = 125$  (corresponding)  
 26 12 apart, half is 6 · 6 right of -9:  $-3$  <sup>1.02</sup>  
 27 No. Corresponding angles on parallel lines are equal.  
 28 yes <sup>2.04</sup>  
 29  $2x + 25 = 61$  ·  $2x = 36$  ·  $x = 18$

30  $y = 180 - 140 = 40$  (straight) ·  $x = 40$  (corresponding)  
 31  $y = 180 - 85 = 95$  (straight) ·  $x = 95$  (corresponding)  
 32  $3x + 5 = 74$  ·  $3x = 69$  ·  $x = 23$   
 33  $75 - 86 = -11$  dollars <sup>1.10</sup>  
 34  $x + 5 = 19$  ·  $x = 14$   
 35  $\angle VXH$  <sup>2.01</sup>  
 36  $y = 180 - 125 = 55$  (straight) ·  $x = 55$  (corresponding)  
 37  $4x + 15 = 39$  ·  $4x = 24$  ·  $x = 6$   
 38  $ZK \parallel HX$  <sup>2.06</sup>

## 2.08 Alternate angles (booklet p. 17)

### Examples (model these live)

E1 The lines are parallel. Find a.  
 alternate angles are equal ·  $a = 50$

E2 a is alternate to  $115^\circ$ . Find a, then b next to it on the line.  
 $a = 115$  (alternate) ·  $b = 180 - 115 = 65$

E3 Alternate angles are  $(3x - 15)^\circ$  and  $60^\circ$ . Find x.  
 $3x - 15 = 60$  ·  $3x = 75$  ·  $x = 25$

### Questions (29 on today's skill, 8 from earlier lessons)

1  $a = 135$   
 2  $106^\circ$   
 3 between them  
 4  $a = 125$   
 5  $49^\circ$   
 6 Z (or N)  
 7  $32^\circ$   
 8 opposite sides  
 9  $a = 95$   
 10  $70 - 79 = -9$  dollars <sup>1.10</sup>  
 11  $a = 155$  (alternate) ·  $b = 180 - 155 = 25$   
 12  $88^\circ$

13  $a = 125$  (alternate) ·  $b = 180 - 125 = 55$   
 14  $a = 120$  · alternate angles,  $\parallel$  lines  
 15 small:  $\angle EVY, \angle YVW$  · whole:  $\angle EVW$  <sup>2.01</sup>  
 16  $x - 15 = 12$  ·  $x = 27$   
 17  $a = 30$  (alternate) ·  $b = 180 - 30 = 150$   
 18  $a = 50$  · alternate angles,  $\parallel$  lines  
 19  $x - 25 = 0$  ·  $x = 25$   
 20 30 <sup>1.01</sup>

21  $a = 140$  (alternate) ·  $b = 180 - 140 = 40$   
 22  $360^\circ - 275^\circ = 85^\circ$  <sup>2.02</sup>  
 23  $3x - 15 = 30$  ·  $3x = 45$  ·  $x = 15$   
 24  $a = 130$  · alternate angles,  $\parallel$  lines  
 25  $2x - 25 = 29$  ·  $2x = 54$  ·  $x = 27$   
 26  $a = 35$  · corresponding  
 27  $2 - (-3) = 2 + 3 = 5^\circ$  <sup>1.05</sup>  
 28  $p = 135$  (alternate) ·  $q = 180 - 135 = 45$  (straight)

29  $c = 360 - 65 - 40$  ·  $c = 255$  <sup>2.05</sup>  
 30  $2x - 5 = 35$  ·  $2x = 40$  ·  $x = 20$   
 31  $a = 30$  · alternate  
 32  $p = 85$  (alternate) ·  $q = 180 - 85 = 95$  (straight)  
 33  $-5$  <sup>1.04</sup>  
 34  $4x - 5 = 107$  ·  $4x = 112$  ·  $x = 28$   
 35  $38^\circ$  <sup>2.04</sup>  
 36  $x - 30 = 7$  ·  $x = 37$   
 37  $a = 45$  · alternate

## 2.09 Co-interior angles (booklet p. 19)

### Examples (model these live)

E1 The lines are parallel. Find a.  
 $a + 110 = 180$  (co-interior) ·  $a = 70$

E2 a is co-interior to  $65^\circ$ . Find a.  
 $a + 65 = 180$  ·  $a = 115$

E3 Co-interior angles are  $y^\circ$  and  $2y^\circ$ . Find y.  
 $y + 2y = 180$  ·  $3y = 180$  ·  $y = 60$

### Questions (30 on today's skill, 11 from earlier lessons)

1  $a = 55$   
 2  $180^\circ - 83^\circ = 97^\circ$   
 3 add to  $180^\circ$

4  $a = 75$   
 5  $180^\circ - 124^\circ = 56^\circ$   
 6 yes

7  $180^\circ - 148^\circ = 32^\circ$   
 8 C (or U)  
 9  $a = 110$

10  $-7$  <sup>1.02</sup>  
 11  $a + 128 = 180$  ·  $a = 52$   
 12  $180^\circ - 34^\circ = 146^\circ$

- 13  $a + 158 = 180 \cdot a = 22$   
 14  $a = 180 - 125 = 55 \cdot$  co-interior angles,  $\parallel$  lines  
 15 small:  $\angle LDG, \angle GDA \cdot$  whole:  $\angle LDA$  2.01  
 16 Yes.  $102 + 78 = 180$ .  
 17  $a + 47 = 180 \cdot a = 133$   
 18  $a = 180 - 55 = 125 \cdot$  co-interior angles,  $\parallel$  lines  
 19 No.  $32 + 138 = 170$ , not 180.  
 20  $-1 \ 518$  1.09  
 21  $a + 39 = 180 \cdot a = 141$   
 22  $360^\circ - 230^\circ \cdot = 130^\circ$  2.02  
 23  $y + 5y = 180 \cdot 6y = 180$ , so  $y = 30 \cdot$  angles:  $30^\circ$  and  $150^\circ$   
 24  $a = 180 - 100 = 80 \cdot$  co-interior angles,  $\parallel$  lines  
 25 opposite sides 2.08  
 26  $y + 3y = 180 \cdot 4y = 180$ , so  $y = 45 \cdot$  angles:  $45^\circ$  and  $135^\circ$   
 27  $2x + 20 + 118 = 180 \cdot 2x = 42 \cdot x = 21$   
 28  $= 2 - (-1) \cdot = 3$  1.08  
 29  $a = 125 \cdot$  corresponding  
 30 SZ is parallel to NR 2.06  
 31  $y + 4y = 180 \cdot 5y = 180$ , so  $y = 36 \cdot$  angles:  $36^\circ$  and  $144^\circ$   
 32  $2x + 30 + 106 = 180 \cdot 2x = 44 \cdot x = 22$   
 33  $1 \times 30^\circ = 30^\circ \cdot$  acute 2.03  
 34  $2x + 20 + 126 = 180 \cdot 2x = 34 \cdot x = 17$   
 35  $a = 50 \cdot$  co-interior  
 36  $25$  1.06  
 37  $y + 8y = 180 \cdot 9y = 180$ , so  $y = 20 \cdot$  angles:  $20^\circ$  and  $160^\circ$   
 38  $44^\circ$  2.04  
 39 No.  $45 + 125 = 170$ , not 180.  
 40  $2x + 10 + 148 = 180 \cdot 2x = 22 \cdot x = 11$   
 41  $a = 105$  2.07

## 2.10 Finding unknown angles on parallel lines (booklet p. 21)

### Examples (model these live)

- E1 Which rule? Find a.  
 Z shape: alternate  $\cdot a = 65$   
 E2 Find b. Give a reason.  
 C shape: co-interior  $\cdot b = 180 - 65 = 115$   
 E3 Find y, then x. Give reasons.  
 $y = 180 - 50 = 130$  (straight)  $\cdot x = 130$  (corresponding)

### Questions (26 on today's skill, 8 from earlier lessons)

- 1 corresponding  
 2  $a = 100$   
 3 are equal  
 4 alternate  
 5  $a = 150$   
 6 are equal  
 7 co-interior  
 8  $a = 95$   
 9 add to  $180^\circ$   
 10 corresponding  
 11  $12$  1.05  
 12  $a = 180 - 100 = 80 \cdot$  co-interior angles add to  $180^\circ$   
 13  $a = 60 \cdot$  corresponding angles are equal  
 14  $y = 180 - 140 = 40$  (straight)  $\cdot x = 40$  (corresponding)  
 15 small:  $\angle KZS, \angle SZX \cdot$  whole:  $\angle KZX$  2.01  
 16  $m + 80 = 180 \cdot m = 100$   
 17  $y = 180 - 65 = 115$  (straight)  $\cdot x = 115$  (corresponding)  
 18  $m = 53 \cdot m = 53$   
 19  $44 - 60 \cdot = -16$  dollars 1.10  
 20  $a = 80 \cdot$  alternate angles are equal  
 21  $360^\circ - 240^\circ \cdot = 120^\circ$  2.02  
 22  $p = 85$  (corresponding, then vert. opp.)  $\cdot q = 180 - 85 = 95$  (straight line)  
 23  $p = 130$  (corresponding, then vert. opp.)  $\cdot q = 180 - 130 = 50$  (straight line)  
 24  $3x + 15 = 75 \cdot 3x = 60 \cdot x = 20$   
 25 total:  $-10 \cdot = -10 \div 5 \cdot = -2^\circ$  C 1.07  
 26 No. Co-interior angles add to  $180^\circ$ , so  $a = 100$ .  
 27  $q$  2.07  
 28  $3x + 15 = 60 \cdot 3x = 45 \cdot x = 15$   
 29 No. Co-interior angles add to  $180^\circ$ , so  $a = 90$ .  
 30  $-6$  1.03  
 31  $p = 40$  (corresponding, then vert. opp.)  $\cdot q = 180 - 40 = 140$  (straight line)  
 32  $131^\circ$  2.04  
 33  $m + 159 = 180 \cdot m = 21$   
 34  $3x + 15 = 111 \cdot 3x = 96 \cdot x = 32$

## 2.11 Proving lines are parallel (booklet p. 23)

### Examples (model these live)

- E1 Alternate angles are  $64^\circ$  and  $64^\circ$ .  
 Is  $AB \parallel CD$ ?  
 $64 = 64 \cdot$  yes: alternate angles are equal  
 E2 Co-interior angles are  $105^\circ$  and  $85^\circ$ . Is  $AB \parallel CD$ ?  
 $105 + 85 = 190 \cdot$  no: they do not add to  $180^\circ$   
 E3 Co-interior angles are  $2x^\circ$  and  $120^\circ$ . Find  $x$  so that  $AB \parallel CD$ .  
 $2x + 120 = 180 \cdot 2x = 60 \cdot x = 30$

### Questions (34 on today's skill, 13 from earlier lessons)

- 1 yes  
 2 they are equal  
 3 yes  
 4 they are equal  
 5 yes  
 6 they add to  $180^\circ$   
 7 yes  
 8 yes  
 9  $a = 90$  2.10  
 10 yes  
 11  $60 = 60 \cdot$  yes: corresponding angles are equal  
 12 yes  
 13  $-5$  1.03  
 14  $100 = 100 \cdot$  yes: alternate angles are equal  
 15 yes  
 16  $125 + 55 = 180 \cdot$  yes: co-interior angles add to  $180^\circ$   
 17  $180^\circ - 36^\circ \cdot = 144^\circ$  2.09  
 18  $30 = 30 \cdot$  yes: corresponding angles are equal  
 19  $2x + 134 = 180 \cdot 2x = 46 \cdot x = 23$   
 20 small:  $\angle TWY, \angle YWV \cdot$  whole:  $\angle TWV$  2.01  
 21  $30 = 30 \cdot$  yes: alternate angles are equal  
 22  $2x + 130 = 180 \cdot 2x = 50 \cdot x = 25$   
 23  $2x + 148 = 180 \cdot 2x = 32 \cdot x = 16$   
 24  $75 = 75 \cdot$  yes: corresponding angles are equal  
 25  $-28$  1.01  
 26  $3x + 10 = 88 \cdot 3x = 78 \cdot x = 26$   
 27  $360^\circ - 210^\circ \cdot = 150^\circ$  2.02  
 28  $2x + 122 = 180 \cdot 2x = 58 \cdot x = 29$   
 29  $80 \neq 85 \cdot$  no: they are not equal  
 30 are equal 2.10  
 31  $3x + 10 = 34 \cdot 3x = 24 \cdot x = 8$   
 32  $3x + 10 = 58 \cdot 3x = 48 \cdot x = 16$   
 33 No.  $70 + 115 = 185$ , not 180.  
 34  $= -12 - 5 \cdot = -17$  1.08  
 35 No. Co-interior angles must add to  $180^\circ$ ;  $125 + 125 = 250$ .  
 36 Z (or N) 2.08  
 37  $3x + 10 = 85 \cdot 3x = 75 \cdot x = 25$   
 38 No.  $80 + 105 = 185$ , not 180.  
 39  $5 \times 30^\circ = 150^\circ \cdot$  obtuse 2.03  
 40 No.  $60 + 125 = 185$ , not 180.  
 41 No. Co-interior angles must add to  $180^\circ$ ;  $115 + 115 = 230$ .  
 42  $2$  1.02  
 43  $3x + 10 = 40 \cdot 3x = 30 \cdot x = 10$   
 44  $163^\circ$  2.04  
 45  $2x + 146 = 180 \cdot 2x = 34 \cdot x = 17$   
 46 No.  $55 + 130 = 185$ , not 180.  
 47  $a = 55$  2.09